

8 - Puzzle Problem

Dfs (start, goal)

create an empty STACK

create an empty VISITED set

push start into STACK

while STACK is not empty:

pop the top state

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if state == goal:

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return solution

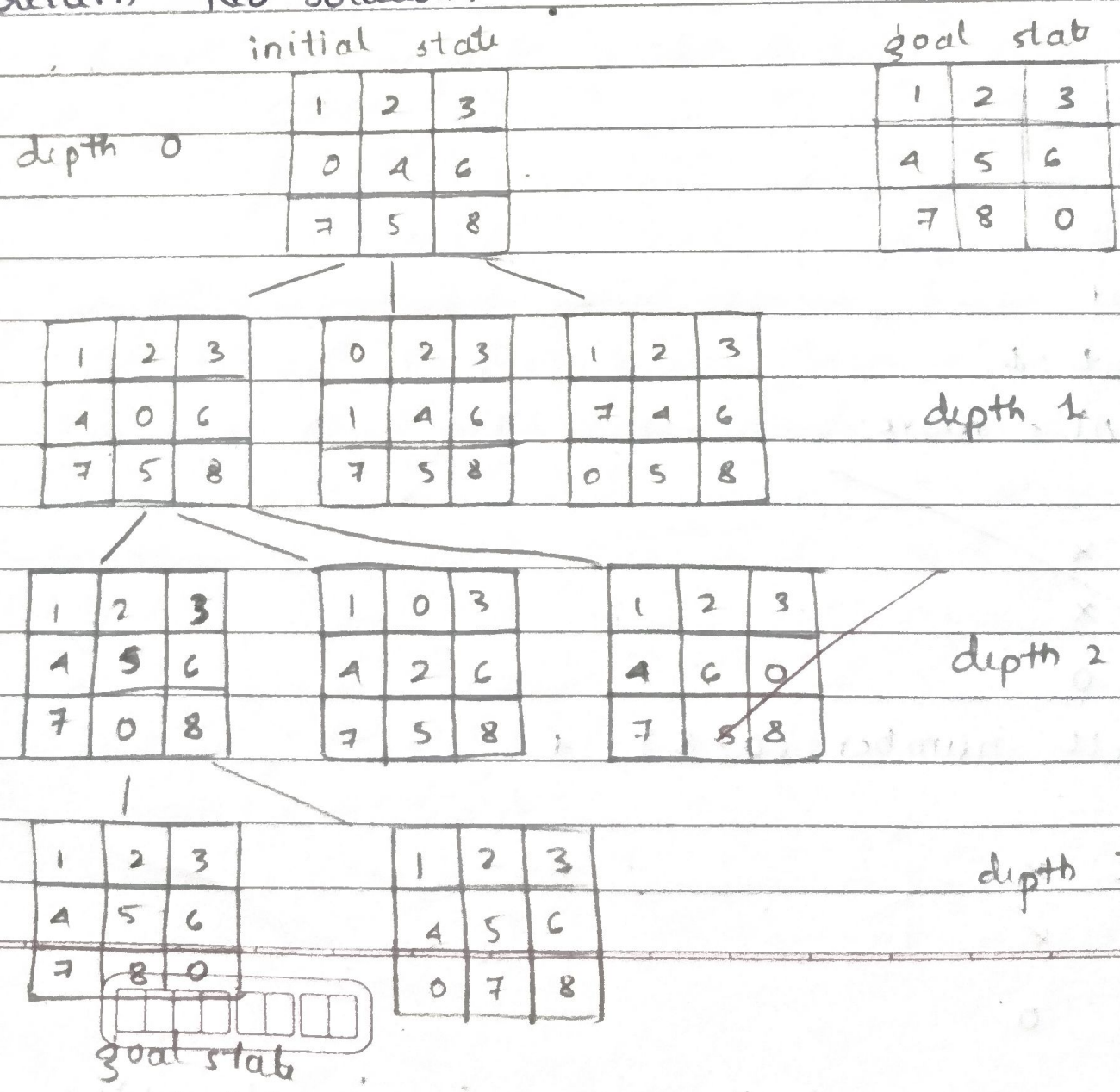
if state is not visited:

mark state as visited

generate all valid moves

push each unvisited child into STACK

return "No Solution".



output

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--- 8 - Puzzle DFS Solver ---

Use '0' to represent the blank space

Enter the INITIAL state:

Enter 9 numbers (0-8) separated by spaces

1 2 3 0 4 6 7 5 8

Enter 9 numbers (0-8) separated by spaces

1 2 3 4 5 6 7 8 0

Initial state grid:

1	2	3
0	4	6
7	5	8

Step 1: Move blank space 'right'

1	2	3
4	0	6
7	5	8

Step 2: Move blank space 'Down'

1	2	3
4	5	6
7	0	8

Step 3: Move blank space 'Right'

1	2	3
4	5	6
7	8	0

Success! Optimal solution found with a path cost of: 3

8/0/0/0/0/0/0/0/0

8	0	0	0	0	0	0	0	0
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